Index Number : ..... Program: .....

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

SCHOOL OF SCIENCES

DEPARTMENT OF MATHEMATICS AND STATISTICS

Level 100, Mid - Semester Examinations , 2016/2017

B.Sc. (Pet./Ren. En./Elect./ Mech. /Env. Eng.),

MATH 109: ENGINEERING MATHS 1 Semester I

11th November, 2016 Time: 1.5 Hours

SECTION A (60% Marks)

1. Solve the inequality $|x+3a| > 2|x-2a|$, a is a positive constant. [4 marks]

2. a) Find the period of $cos\left(\frac{2x-1}{5}\right)$ [2 marks]

b) Decide whether the functions given by

i. $f(x) = x^3 tanx$ ii. $f(x) = sin2x - sinx$

is either odd, even or neither. [3 marks]

4. If ϕ is an empty set, complete the sentence using the following

$\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$

$\phi \subset$ ..... $\subset$ .....$\subset$.....$\subset$..... [2 marks]

5. Complete the following identities in hyperbolic functions

a) $cosh^2x - sinh^2x \equiv$ ..... [0.5marks]

b) $cosech^2x \equiv$ ..... [0.5marks]

c) $cosh2x \equiv$ ..... [0.5marks]

d) $tanh2x \equiv$ ..... [0.5marks]

6. Complete by simplifying the following

a) $\left|\frac{x}{y}\right| =$ .... [1 marks]

b) $|x+y|$ ..... [1 marks]

c) $|x_1 \times x_2 \times \ldots\ldots\ldots \times x_n| =$ ..... [1 mark]

d) $|x|^2 \equiv$ ..... [1 mark]

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e) $|x| = |y|, \Rightarrow$ ..... [1 mark]

7. Give 3 **different** examples each of the following.

a) Rational Numbers :

..... [1 mark]

b) Irrational Number :

..... [1 mark]

c) Complex Numbers :

.....[1 mark]

d) Integers:

..... [1 mark]

8. Evaluate

a) $\lim_{x \to 2} \frac{\sqrt{x^2+5}-3}{x^2-2x}$ b) $\lim_{x \to 2} \frac{\sqrt{x+4}-2}{x}$ [4 marks]

9. Find the value of c for which the function

$f(x) = \begin{cases} \frac{x^2-16}{x-4}, & x \neq 4 \\ C, & x = 4 \end{cases}$ is continuous [3 marks]

10. Does the $\lim_{x \to 2} f(x)$ exists for $f(x) = \begin{cases} 7x-2, & if\ x \geq 2 \\ 3x+5, & if\ x < 2 \end{cases}$ [4 marks]

11. Differentiate the following with respect to x.

a) $In(sin2x)$ b) $cos^2(4x+3)$ c) $tan(xy) = y$ [6 marks]

ANSWER ONLY ONE QUESTION

SECTION B (40% Marks)

12. Prove by induction that for all positive integral values of n,

$$\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$ [7 marks]

13. The complex number $2+i$ is denoted by u.

b) Given that u is a root of the equation $x^3 - 11x - k = 0$, where k is real. Find the value of k. [3 marks]

c) Write down the other complex root of the equation. [1 mark]

d) Find the modulus and argument of u. [3 marks]

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DEPARTMENT OF MATHEMATICS AND STATISTICS

Mid-Semester Examination , 2019/2020

B.Sc. (Pet./Ren. En./Elect./ Mech. /Env. Eng./ Agric./Comp. Eng.)

Candidates Programme: _____

Index Number: | | | | | | | | | | |

ENGINEERING MATHS I — MATH 109

LEVEL 100 — 2.00 HRS

Additional Materials: Non-programmable Calculator — 08TH NOVEMBER, 2019

SECTION A (80%)

1) Solve the equations

a) $\frac{|x|+1}{|x|-1} < 4$

b) $z^2 + 2\sqrt{3}iz - 4 = 0$

c) $2\tanh^{-1} x = \ln 3$

[3 marks each]

2) Differentiate

a). $In\left(\frac{x}{y}\right)$

c) $y = \tan^{-1}\left(\frac{a+x}{1-ax}\right)$

3) Prove that

a) $b < \frac{ae^{x}+be^{-x}}{e^{x}+e^{-x}} < a$ simplify to $a > b > 0$

b) $cosh2x = 1 + 2\sinh^2 x$

4. Find:

a) $\lim_{x\to 1} \frac{x^3-x^2+x-1}{x+lnx-1}$

b) $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{sinx}\right)$

5. The curve C: $3^{x-1} + xy - y^2 + 5 = 0$. Find $\frac{dy}{dx}$ at points (1,3). [4marks]

6. a) Find the possible values of x in $\begin{vmatrix} coshx & -sinhx \\ sinhx & coshx \end{vmatrix} = 2$ [4 marks]

b) Let $f(z) = z^4 - a^4$ where $f(z) \in \mathbb{C}$. Factorize $f(z)$ completely.

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7. Discuss the continuity of the function

$$f(x) = \begin{cases} \frac{x^3}{a} - a^3, & 0 < x < a \\ 0 & x = a \\ a - \frac{a^2}{x}, & x > a \end{cases}$$

8. Verify Rolle's theorem for:

$$f(x) = \ln\left(\frac{x^2+3}{4x}\right) \text{ on } [1,3]$$

9. The complex numbers α and β are given by: $\frac{\alpha+4}{\alpha} = 2 - i$ and $\beta = -\sqrt{6} + \sqrt{2}i$

a) Show that $i)$ $\alpha = 2 + 2i$ [3marks] $ii)$ $|\alpha| = |\beta|$.

b) Find the modulus and argument of $\alpha\beta$

10. Given $z = cos\theta + isin\theta$. Find:

a) $|z + 1|$ b) $\arg(z + 1)$

SECTION B (20%)

Answer Only One Question

1. The complex number z is defined by $z = a + ib$, where a and b are real. The complex c is denoted by z^*.

(i) Show that $|z|^2 = zz^*$ and that $(z - ki)^* = z^* + ki$, where k is real.

In an Argand diagram a set of points representing complex numbers z is defined by the equation $|z - 10i| = 2|z - 4i|$.

(ii) Show, by squaring both sides, that

$$zz^* - 2iz^* + 2iz - 12 = 0.$$

Hence show that $|z - 2i| = 4$. [5]

(iii) Describe the set of points geometrically. [1]

2. a) It is given that $\frac{d^n}{dx^n}\left(\frac{lnx}{x}\right) = \frac{a_n lnx + b_n}{x^{n+1}}$ where a_n and b_n depend only on n.

i) Find a_1, a_2 and a_3. [6 marks]

ii) Use the mathematical induction to establish a formula for a_n. [4 marks]

b) Prove by induction that $5^n + 12n - 1$ is divisible by 16. [5 marks]

c) Prove that $\sqrt{5}$ is irrational. [4 marks]

$|x-y| = |y-x|$

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ENGINEERING MATHS I — MATH 109

LEVEL 100 — 1HR.30MINS

Additional Materials: Non-programmable Calculator — 19TH MARCH, 2021

ANSWER ALL QUESTIONS

SECTION A (70%)

1. Find the following limits if they exists

a) $\lim_{x\to 2}\left(\frac{1}{x-2}-\frac{4}{x^2-4}\right)$ b) $\lim_{x\to -\infty}\left(\frac{4x-3}{\sqrt[3]{x^2+1}}\right)$ c) $\lim_{x\to +\infty}\left(\frac{7x-4}{\sqrt{x^3+5}}\right)$ [2 marks]

2. For what value of k is the function continuous?

$$f(x)=\begin{cases}\dfrac{\sqrt{7x+2}-\sqrt{6x+4}}{x-2}, & if\ x\geq \dfrac{-2}{7}\ and\ x\neq 2\\ k, & if\ x=2\end{cases}$$

3. Evaluate the limits (if its exists)

$$\lim_{x\to 2} f(x),\ \text{if}\ f(x)=\begin{cases}7x-2, & if\ x\geq +2\\ 3x+5, & if\ x<2\end{cases}$$

4. By solving the inequalities, write your answer in the form of interval (bracket type).

a) $\frac{(2x-1)(x+1)}{(3-x)}<0$ [2 marks] b) $\frac{3x^2-12x-4}{x^2-4x+9}>0$ [2 marks]

$(2x-1)(x+1)<0$ $3x^2-12x-4<0$

$2x+1<0$ $x+1<0$

$x=\frac{1}{2}$ $x=-1$

$-1<x<\frac{1}{2}$

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5. A sequence of numbers $u_1, u_2, u_3, u_4, \ldots$ is defined by

$$u_{n+1} = 4u_n + 2, \qquad where \;\; u_1 = 2$$

Prove by induction that, for $n \in \mathbb{Z}^+$, $u_n = \frac{2}{3}(4^n - 1)$ [7 marks]

6. a) By considering $z = cos\theta + isin\theta$, simplify $\sum_{r=1}^{n} z^{2r-1}$ if $sin\theta \neq 0$. [3 marks]

(Hint. De' Moivres theorem will be useful)

b) Solve the inequality $|x + 3a| > 2|x - 2a|$, a is a positive constant. [4 marks]

$(x+3a)^2 - (2x-2a)^2$

$x+3a+2x-5$

7. Given that 2 and $5 + 2i$ are roots of the equation

$$x^3 - 12x^2 + cx + d = 0, \quad \text{for } c, d \in \mathbb{R}$$

a) Find the other roots

b) Find the values of c and d

$z^{2r-1} = (\cos\theta + i\sin\theta)^{2r-1}$

$z^n = \cos n\theta + i\sin n\theta$

8. State

True or false?

a $\mathbb{Z}^+ \subseteq \mathbb{N}$	b $\mathbb{N} \subset \mathbb{Z}$	c $\mathbb{N} = \mathbb{Z}^+$	d $\mathbb{Z}^- \subseteq \mathbb{Z}$
e $\mathbb{Q} \subset \mathbb{Z}$	f $\{0\} \subseteq \mathbb{Z}$	g $\mathbb{Z} \subseteq \mathbb{Q}$	h $\mathbb{Z}^+ \cup \mathbb{Z}^- = \mathbb{Z}$

SECTION B (30%)

Answer Only One Question

$z^3 + 4\sqrt{3} = 4i$

$(x+iy)^3 + 4\sqrt{3} = 4i$

9. Prove by mathematical induction that

$\frac{2x}{4}$ $\quad u = 2n$ $\quad x = 4$

$$\sum_{r=1}^{n} r.In\left(\frac{r+1}{r}\right) = In\left[\frac{(n+1)^n}{n!}\right] \text{ For all } n \epsilon \mathbb{Z}^+.$$ [6marks]

10. Find the roots of the equation $z^3 + 4\sqrt{3} = 4i$, giving your answers in the form $re^{-i\theta}$ where $r > 0$ and $0 < \theta < 2\pi$ [7 marks]

11. Two complex numbers w and z are defined by $w = 5 + 3i$ and $z = 4 + i$

a) Express $\frac{iw}{z}$ in the form $x + iy$, showing all your working and giving the exact values of x and y. [3marks]

b) Find wz and hence by considering arguments, show that

$tan^{-1}\left(\frac{3}{5}\right) + tan^{-1}\left(\frac{1}{4}\right) = \frac{\pi}{4}$. [4 marks]

θ-

$(x+iy)(x-iy)$

$x^2 + ixy + ixy + i^2y^2$

$x^2 + 2ixy - y^2$